

ARTIFICIAL INTELLIGENCE IN PUBLIC HEALTH: LESSONS FROM THE EUROPEAN PUBLIC HEALTH CONFERENCE

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> #EUPHA2024 – Plenary 1:

Navigating the Artificial Intelligence (AI) wave: overcoming barriers and unleashing the potential of AI in transforming European public health

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Summary: Artificial intelligence (AI) holds great promise for advancing public health functions. This article, drawing on discussions from Plenary 1 at the European Public Health Conference 2024, considers some ways in which AI is already used in public health, including for health promotion, surveillance and protection, population health assessment, disease and injury prevention, emergency prediction, preparedness and response, and resource optimisation. It also explores what actions need to be taken in the future to help ensure AI is meaningfully embedded in public health practice, and to overcome key challenges such as ensuring equity, avoiding biases, maintaining trust, and safeguarding privacy.

Keywords: Artificial Intelligence, Digital Health, Regulation, Public Health, Inequalities

Introduction

Interest in applying Artificial Intelligence (AI) in health has grown rapidly in the past five years. AI's ability to quickly analyse and extract large volumes of diverse and complex data, offer personalised recommendations, provide decision support, and enhance operational efficiency has been hailed as a game-changer for the health sector. While much of the discussion around AI in health centres on its potential benefits and risks for patients and healthcare professionals in healthcare delivery, AI also offers considerable promise for advancing public health functions.^{1–3} These include health promotion, surveillance and protection,

population health assessment, disease and injury prevention, emergency prediction, and preparedness and response. Table 1 highlights how various branches of AI can contribute to public health and healthcare.

At the same time, implementing AI in healthcare comes with significant challenges, such as data privacy concerns, ensuring data quality and representativeness, unclear accountability, difficulties integrating AI tools into existing systems, the risk of exacerbating inequalities, and the need for both health professionals and the public to acquire the necessary skills to use AI effectively.^{1–3}

Box 1: Integrating Generative AI in a conference plenary – an insight into the workflow

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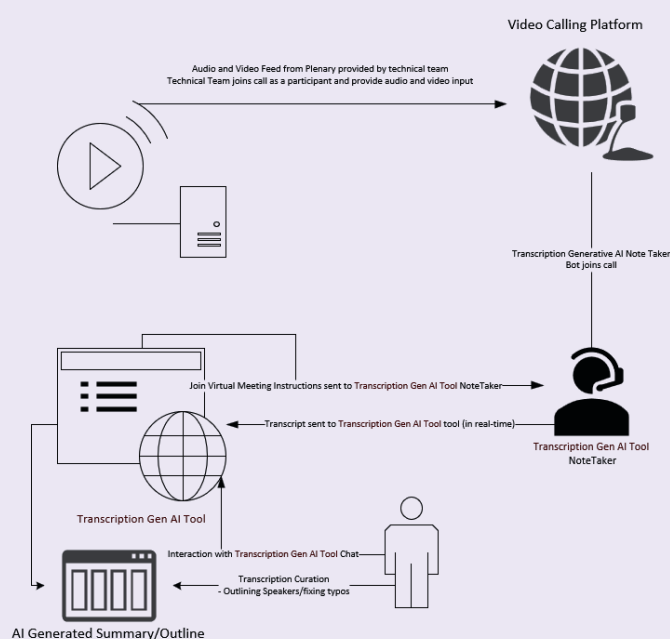
This workflow depicts the unique setup to create a real-time AI-generated summary of the plenary session on AI in public health at the 17th European Public Health Conference. This session also explored the concept of an “AI-enabled panellist” by interacting with AI tools to simulate real-time contributions to the discussion. The aim was to enhance knowledge-sharing, create structured outputs for post-event use, and demonstrate a practical and current use of generative AI for conference participants.

- **Audio and Video Feed Input:** The conference technical team provided a live audio and video feed from the plenary session. This feed was transmitted into cloud-based team collaboration software, where the session was hosted as a virtual meeting.
- **Integration with AI-based text transcription:** an AI text transcription bot was invited as a participant in the online meeting. This bot captured the plenary discussion in real-time, producing:
 - Real-Time Transcription, which was made available to participants and administrators during the session for immediate reference.
 - AI-Enabled Panellist Experimentation: the AI transcription bot was prompted by the facilitator during the session to provide insights, summaries or clarifications based on the ongoing discussion.
- **Curating the Transcription:** The automated transcript was reviewed and refined to ensure accuracy and usability following the session. This curation process involved:
 - Labelling speakers to improve clarity.
 - Correcting typographical errors.
 - Highlighting key points and synthesising an AI-generated summary.

- **AI-Driven Summarisation and Output:** The refined transcript was processed to create a structured summary of the plenary discussion. Leveraging AI chat capabilities, the summarisation process ensured that the outputs were relevant, comprehensive, and aligned with the key themes of the session.
- **Impact and Utilisation:** The final outputs, including curated transcripts and AI-generated summaries, were shared with stakeholders as a session record.
- **Backup Options:** The setup also allowed the plenary organiser to use the premium features within the team collaboration software for real-time transcription. As a third and final backup, the facilitator used a smartphone device as an additional input device for the AI transcription bot.

This workflow highlights how AI tools can be integrated into public health events to document proceedings and facilitate innovative interactions, such as using AI as a virtual contributor.

Figure 1: Workflow of Setup to enable AI-generated transcription and summary during the EPH Plenary on AI in Public Health



To discuss how AI can be meaningfully embedded in public health practice and how the public health community can reap the rewards while minimising risks, a plenary session was convened at the 17th European Public Health Conference held in Lisbon in November 2024. It brought together key stakeholders from public health institutions, academia and international

organisations and leveraged audience participation to explore how AI in and for public health is currently perceived and what important steps can be taken in the future. It also used AI to capture insights from both panellists and the audience in real-time (see Box 1).

This article presents key insights from the plenary session. Based on input from the audience, it starts by looking at the current use of AI among public health professionals in general, followed by the specialised applications developed at a specialist centre. It then reflects on key themes from the panel discussion, focusing on the main areas for future action.

Table 1: Common terms used in AI and their application in public health and healthcare.

BRANCH OF AI	EXAMPLE OF USES IN PUBLIC HEALTH	EXAMPLE OF USES IN HEALTHCARE
Expert systems AI programs that mimic the decision-making abilities of humans use set rules and logic to provide recommendations	Expert systems help public health officials determine vaccination strategies by analysing population data, vaccination rates, and outbreak patterns	Expert systems like MYCIN (originally used to diagnose bacterial infections) have evolved to assist in diagnosing various diseases based on patient symptoms and test results
Machine Learning (ML) Algorithms that enable computers to learn from and make predictions based on data	Algorithms analyse historical health data to predict outbreaks of diseases like influenza or COVID-19, enabling proactive responses	Models assess individual risk factors for chronic diseases by analysing demographic, lifestyle, and clinical data, helping in targeted interventions
Reinforcement Learning A type of ML where an agent learns to make decisions within a given environment by receiving penalties and rewards	Modelling to improve public health campaigns by simulating various intervention strategies and their effects on community behaviour	Algorithms optimise the allocation of healthcare resources during crises, like distributing vaccines or medical supplies based on real-time needs
Foundational Model¹⁵ Foundational Model (FM) is a type of AI model that is trained on a diverse dataset that can include text, images, speech, structured data and 3D Signals to learn general patterns and relationships allowing it to be fine-tuned for a wide range of specific tasks and applications	FM can be used to predict diagnosis-related groups for hospitalised patients ¹⁶	FM can support the development of generalist medical AI for bedside decision support, radiology reporting and augmented procedures
Natural Language Processing (NLP) Focuses on the interaction between computers and human language, enabling machines to understand, interpret and generate text/speech	NLP tools analyse social media and public comments to gauge public sentiment regarding health campaigns or vaccination drives, informing communication strategies	NLP is used to extract valuable insights from unstructured clinical notes in electronic health records, facilitating better patient management
Large Language Model Large Language Model (LLM) is a type of AI technology trained on vast amounts of text data to generate human-like language, allowing it to understand, analyse, and respond to complex queries	LLMs can be used to summarise research on a public health topic, allowing for a quicker turnaround in gathering evidence for specific public health policy decisions	LLMs can be deployed in a healthcare setting to support clinicians with clinical documentation (emulating the role of medical scribes) and creating discharge summaries
Computer Vision Enables machines to interpret and understand visual information and make decisions based on this information	Computer vision algorithms analyse satellite imagery to monitor environmental factors affecting public health, such as air quality and land use	AI systems process images from X-rays, MRIs, or CT scans to assist in early detection of diseases, such as tuberculosis or cancer, improving diagnostic efficiency
Robotics Design of robots that can perform tasks autonomously or semi-autonomously using AI for navigation, manipulation and decision-making	Autonomous drones and ground robots collect health data in hard-to-reach areas, such as remote villages or during natural disasters	Robots can be deployed for the delivery of medications and supplies in healthcare settings, reducing human contact and improving efficiency, especially during pandemics

Note: These distinctions are not always absolute; for instance, computer vision and deep learning (a type of machine learning that uses neural networks with multiple layers to learn from big data) are often combined for diagnostic purposes in medical imaging and pathology, and foundational models often build on large language models (LLMs).

for people without private vehicles, especially in smaller cities and rural regions.⁹

- **Enhancing Public Health Research Methods:** The Centre also focuses on improving the quality and efficiency of public health research by optimising methods. For example, researchers are exploring and advancing the capabilities of natural language processing tools (see Table 1) to feed into the preparation and execution of systematic reviews, for example in the area of immunisation.¹⁰

The projects in the Centre's PhD programme's first three cohorts can be viewed on the RKI website.¹¹ The programme admits new students annually, with calls for applications for the next cohort to open in spring 2025.

Learning from different perspectives – insights from the panel discussion

Drawing on the insights provided by the audience and the example of the RKI Centre on AI in Public Health Research, the panel discussion revolved around key

considerations for future actions to build on the potential of AI for public health. AI itself (See Figure 1 in Box 1) identified the following themes as they emerged from the panel discussion:

- Establishing transparent **governance and regulatory frameworks** to guide responsible AI development and deployment. This was considered key to ensuring that AI in health brings the intended benefits and does not expose people and the health systems to unnecessary risks. Consequently, it is

Box 2: The EU's role in advancing AI in health

- **EU AI Act:** The EU introduced the EU AI Act in 2024, creating an overarching regulatory framework to ensure the ethical development, safety, and protection of health and fundamental rights when using AI in healthcare.
- **Data Governance:** The General Data Protection Regulation (GDPR) sets the framework for protecting personal data across the EU, crucial for AI technologies that depend on health data. The Open Data Directive and Data Governance Act also increase data availability and sharing, supporting AI development. Once adopted, the European Health Data Space (EHDS) will facilitate cross-border access to health data, promote secure data sharing and collaboration, and enable the development of AI technologies for public health and healthcare delivery while ensuring privacy and data protection.
- **Technology Regulation:** The Medical Devices Regulation (EU) 2017/745 and the In Vitro Diagnostic Medical Devices Regulation (EU) 2017/746 regulate AI embedded in medical devices, ensuring they meet safety and performance requirements.
- **Supporting Innovation:** The EU promotes AI innovation through funding programmes such as Horizon Europe and allows the testing of AI systems in controlled environments like regulatory sandboxes.

Box 3: Six ethical principles in the use of AI from the WHO's guidance on the ethics and governance of AI for health

AI should not undermine human autonomy. In healthcare, humans should remain in control and users must have the necessary information to use AI systems safely and effectively, with patients understanding the role of AI in their care. This also includes protection of privacy, confidentiality, and valid informed consent.

AI technologies should be designed and used in ways that promote the well-being and safety of individuals and the public. This includes ensuring that AI systems do not cause harm and that they contribute positively to health outcomes.

Transparency is critical for the ethical use of AI. It involves providing sufficient information about AI technologies to enable evaluators and regulators to conduct effective oversight. Transparency also includes making information

about the assumptions, limitations, and development of AI technologies available.

All stakeholders involved in the development and deployment of AI technologies should be held responsible for their actions. This includes ensuring that AI systems are used ethically and that there are mechanisms in place to address any violations of ethical principles.

AI technologies should be designed and used in ways that ensure inclusiveness and equity. This means that all individuals, including vulnerable groups, should benefit from AI technologies, and that these technologies should not exacerbate existing biases or discrimination.

Finally, given the enormous energy requirements, AI systems should be energy-efficient, minimising their environmental impact.

Source:¹⁷

also an important driver for generating trust in AI-driven tools, among health professionals, patients, and the public in general. The European Union (EU) has taken key steps to advance these objectives (see Box 2).¹²

- Addressing **data governance, privacy and equity** concerns. There are several ethical dimensions to consider in developing and implementing AI tools, spanning data protection, the mitigation of algorithmic biases, and ensuring equitable access to safe and effective AI-powered tools. The World Health Organization has developed a set of ethical principles to guide the use of AI, including autonomy, transparency,

responsible use, inclusiveness and environmental sustainability (see Box 3).¹³

- Improving **AI literacy** among public health professionals and the public. Investing in AI literacy can build trust, ensure the best possible use of AI tools and foster accountability and equity, including exploration of the potential of explainable AI.¹⁴ This also means understanding the limitations of what AI can provide, emphasising tasks where it adds value in supporting public health professionals rather than replacing them.¹⁵
- Building the **evidence on the benefits and risks of AI** in public health. To

further support trust and adoption, there is a need for strong evidence about its actual contribution to improving the effectiveness and efficiency of public health practice. Starting with small-scale pilots, rigorously evaluating outcomes, and sharing lessons learned would help build confidence in AI use. The “business case” still needs to be developed for AI tools aiming to provide economic benefits (e.g. by simplifying administrative tasks).

- Fostering **multi-stakeholder collaboration and knowledge sharing**. The discussion underscored the value of bringing together diverse stakeholders, including policymakers, industry, academia, and civil society,

to develop strategies for implementing AI and co-create trusted AI solutions. Creating a platform where countries can share experiences and best practices around implementing AI and align on a regional strategy for AI in public health could help with knowledge transfer and prioritisation for investment.

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Conclusion

The European Public Health Association put AI centre stage by dedicating the first plenary session of its 2024 annual conference to the meaningful use of AI for advancing public health policy and practice. The session reflected both the enthusiasm for what these innovative technologies can achieve for public health and the reservations about their deployment at scale without the necessary guardrails. The need to invest in data systems, evidence generation, fit-for-purpose regulation, and guidance addressing privacy, transparency, accountability, and equity, as well as in AI literacy for health professionals and the public, was highlighted. Fostering multi-stakeholder and international collaboration is key for achieving these goals, and continuing the discussion at fora such as the EPH Conference is an integral part of such efforts.

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